

STAT525 HOMEWORK #3

1. KNNL Problem 2.57.
2. KNNL Problem 2.61.
3. KNNL Problem 3.19 (STAT Majors)
4. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28.
 - (a) Describe the distribution of the explanatory variable.
 - (b) Run the linear regression to predict the county crime rate from the percentage of individuals having at least a high school diploma.
 - (c) Plot the residuals versus the explanatory variable and briefly describe the plot noting any unusual patterns or points.
 - (d) Examine the distribution of the residuals by getting a histogram and a normal probability plot of the residuals by using the HISTOGRAM and QQPLOT statements in PROC UNIVARIATE. What do you conclude?
5. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28. Change the data set by changing the value of the crime rate for the last observation from 7582 to 758 (e.g., a typo). You can do this in a data step. For example,

```
DATA a2; SET a1; IF _n_ EQ 84 THEN y=758;
```

An alternative is to simply edit the data file.

- (a) Make a table comparing the results of this analysis with the results of the analysis of the original data. Include in the table the following: fitted equation, t-test for the slope, with standard error and p -value, R^2 , and the estimate of σ^2 . Briefly summarize the differences.
 - (b) Repeat parts (c) and (d) from the previous problem for this altered data set analysis and summarize how these plots help you to detect the unusual observation.
6. (SAS Exercise) Use the **sales growth** data described in KNNL Problem 3.17.
 - (a) Generate a scatterplot of the data and discuss the appropriateness of using a linear regression model.
 - (b) Using PROC TRANSREG, which power transformation of Y (i.e., value of λ) is most appropriate to use here?
 - (c) Apply this transformation of Y and generate a scatterplot. Again comment on the appropriateness of using a linear regression model.
 - (d) Run the regression model using the transformed data and generate a residual plot (using X or $\hat{Y}$) and a normal probability plot. What do the plots show?
 - (e) Express the estimated regression function in the original units.

1. KNNL Problem 2.57.

2.57. The normal error regression model (2.1) is assumed to be applicable.

a. When testing $H_0: \beta_1 = 5$ versus $H_a: \beta_1 \neq 5$ by means of a general linear test, what is the reduced model? What are the degrees of freedom df_R ?

• Under $H_0, \beta_1 = 5$. We fix the slope at 5 and estimate only β_0 : $Y_i = \beta_0 + 5X_i + \varepsilon_i$

• $Y_i = \beta_0 + 5X_i + \varepsilon_i$ is a model with only one parameter (β_0)

↓

Reduced Model

• $Def: (\text{Parameter in Full}) - (\text{Parameter in Reduced})$

• $Def = 2 - 1 = 1$

b. When testing $H_0: \beta_0 = 2, \beta_1 = 5$ versus H_a : not both $\beta_0 = 2$ and $\beta_1 = 5$ by means of a general linear test, what is the reduced model? What are the degrees of freedom df_R ?

• Under H_0 , both coefficients are fixed: $Y_i = 2 + 5X_i + \varepsilon_i$

• $Y_i = 2 + 5X_i + \varepsilon_i$ is a model with no parameters estimated: fixed line

↓

Reduced Model

• $Def: (\text{Parameter in Full}) - (\text{Parameter in Reduced})$

• $Def = 2 - 0 = 2$

• Full Model:

This model is:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (2.1)$$

where:

β_0 and β_1 are parameters

X_i are known constants

ε_i are independent $N(0, \sigma^2)$

$$b_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} \quad (1.10a)$$

$$SSR = b_1^2 \sum (X_i - \bar{X})^2 \quad (2.51)$$

$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ with two parameters (β_0, β_1)

$$SSTO = \sum (Y_i - \bar{Y})^2$$

$$\frac{SSR}{SSTO} = r^2$$

2. KNNL Problem 2.61.

2.61. Show that the ratio $SSR/SSTO$ is the same whether Y_1 is regressed on Y_2 or Y_2 is regressed on Y_1 . [Hint: Use (1.10a) and (2.51).]

• Regress Y_1 on Y_2 :
(Let $X = Y_2$, $Y = Y_1$):

$$b_1 = \frac{\sum (Y_2 - \bar{Y}_2)(Y_1 - \bar{Y}_1)}{\sum (Y_2 - \bar{Y}_2)^2}$$

$$SSR = b_1^2 \sum (Y_2 - \bar{Y}_2)^2$$

$$SSR = \left(\frac{\sum (Y_2 - \bar{Y}_2)(Y_1 - \bar{Y}_1)}{\sum (Y_2 - \bar{Y}_2)^2} \right)^2 \cdot \sum (Y_2 - \bar{Y}_2)^2$$

$$SSR = \frac{[\sum (Y_2 - \bar{Y}_2)(Y_1 - \bar{Y}_1)]^2}{\sum (Y_2 - \bar{Y}_2)^2}$$

$$SSTO = \sum (Y_1 - \bar{Y}_1)^2$$

$$\frac{SSR}{SSTO} = \frac{[\sum (Y_2 - \bar{Y}_2)(Y_1 - \bar{Y}_1)]^2}{\sum (Y_2 - \bar{Y}_2)^2 \sum (Y_1 - \bar{Y}_1)^2} = r^2$$

$$r = \frac{\sum (Y_2 - \bar{Y}_2)(Y_1 - \bar{Y}_1)}{\sqrt{\sum (Y_2 - \bar{Y}_2)^2 \sum (Y_1 - \bar{Y}_1)^2}}$$

• Regress Y_2 on Y_1 :
(Let $X = Y_1$, $Y = Y_2$):

$$b_1 = \frac{\sum (Y_1 - \bar{Y}_1)(Y_2 - \bar{Y}_2)}{\sum (Y_1 - \bar{Y}_1)^2}$$

$$SSR = b_1^2 \sum (Y_1 - \bar{Y}_1)^2$$

$$SSR = \left(\frac{\sum (Y_1 - \bar{Y}_1)(Y_2 - \bar{Y}_2)}{\sum (Y_1 - \bar{Y}_1)^2} \right)^2 \cdot \sum (Y_1 - \bar{Y}_1)^2$$

$$SSR = \frac{[\sum (Y_1 - \bar{Y}_1)(Y_2 - \bar{Y}_2)]^2}{\sum (Y_1 - \bar{Y}_1)^2}$$

$$SSTO = \sum (Y_2 - \bar{Y}_2)^2$$

$$\frac{SSR}{SSTO} = \frac{[\sum (Y_1 - \bar{Y}_1)(Y_2 - \bar{Y}_2)]^2}{\sum (Y_1 - \bar{Y}_1)^2 \sum (Y_2 - \bar{Y}_2)^2} = r^2$$

$$r = \frac{\sum (Y_1 - \bar{Y}_1)(Y_2 - \bar{Y}_2)}{\sqrt{\sum (Y_1 - \bar{Y}_1)^2 \sum (Y_2 - \bar{Y}_2)^2}}$$

same whether Y_1 is regressed on Y_2 or vice versa.

3. KNNL Problem 3.19

3.19. A student fitted a linear regression function for a class assignment. The student plotted the residuals e_i against Y_i and found a positive relation. When the residuals were plotted against the fitted values $\hat{Y}_i$, the student found no relation. How could this difference arise? Which is the more meaningful plot?

$$e_i = Y_i - \hat{Y}_i \rightarrow Y_i = e_i + \hat{Y}_i$$

- Residuals and Observed values are algebraically dependent.
- If $Y_i \uparrow$, $e_i \uparrow$ (especially when $\hat{Y}_i$ is constant or linear)
- Plotting e_i v.s. Y_i artificially $\uparrow$ correlation, as they're not independent. $\therefore$ fake correlation

- Plotting Residuals and Fitted values have no correlation by construction.
- In OLS Regression, Residuals are uncorrelated with Fitted values: $\sum (e_i \hat{Y}_i) = 0$.
- Plotting e_i v.s. $\hat{Y}_i$ correctly reflects whether model assumptions are satisfied.

Hence, Residuals v.s. Fitted values plot is more meaningful and should be used for model diagnostics.

4. (SAS Exercise) Use the **crime rate** data described in KNNL Problem 1.28.

(a) Describe the distribution of the explanatory variable.

```

1 data crime;
2   input crime_rate hs_diploma_pct;
3   datalines;
4   8487      74
5   8179      82
6   8362      81
7   8220      81
8   6246      87
9   9100      66
10  6561      68
11  5873      81
12  7993      74
13  7932      82
14  6491      75
15  6816      82
16  9639      78
17  7582      76
18 ;
19 run;

```

• Explanatory Variable: **hs_diploma_pct**;

(% of individuals with at least a high school diploma).

• **hs_diploma_pct** is based on 84 observations.

• Mean: 78.60, Median: 79.00, indicating roughly symmetric distribution.

• SD: 6.221, variance: 38.70, showing moderate variability across counties.

• Mode: 82.00, the most frequently occurring value.

• Range: 30.00 (61~91), IQR: 6.5, suggesting the middle 50% of counties fall between 75.0~81.5.

• Skewness of -0.71, indicating distribution is slightly left-skewed.

Meaning there are some counties with lower % high school diploma pulling the distribution leftward.

• Also visible from evidence of extreme values at the lower end: 61, 64, 65.

Distribution of High School Diploma Percentage

The UNIVARIATE Procedure
Variable: hs_diploma_pct

Moments			
N	84	Sum Weights	84
Mean	78.5952381	Sum Observations	6602
Std Deviation	6.22106613	Variance	38.7016638
Skewness	-0.7063786	Kurtosis	0.67561901
Uncorrected SS	522098	Corrected SS	3212.2381
Coeff Variation	7.91532195	Std Error Mean	0.67877396

Basic Statistical Measures			
Location		Variability	
Mean	78.59524	Std Deviation	6.22107
Median	79.00000	Variance	38.70166
Mode	82.00000	Range	30.00000
		Interquartile Range	6.50000

Tests for Location: Mu0=0				
Test	Statistic	p Value		
Student's t	t 115.79	Pr > t	<.0001	
Sign	M 42	Pr >= M	<.0001	
Signed Rank	S 1785	Pr >= S	<.0001	

Quantiles (Definition 5)	
Level	Quantile
100% Max	91.0
99%	91.0
95%	88.0
90%	85.0
75% Q3	82.5
50% Median	79.0
25% Q1	76.0
10%	70.0
5%	65.0
1%	61.0
0% Min	61.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
61	39	88	45
64	76	88	82
64	71	89	29
64	51	91	40
65	20	91	81

(b) Run the linear regression to predict the county crime rate from the percentage of individuals having at least a high school diploma.

```
91 proc reg data=crime;
92     model crime_rate = hs_diploma_pct;
93     title "Linear Regression of Crime Rate on High School Diploma Percentage";
94 run;
```

Regression Equation:

$$\text{crime_rate} = 201518 - 170.58 \text{ hs_diploma_pct};$$

- Slope: -170.58, indicating for each 1% ↑ in high school diploma rate, the predicted crime rate ↓ by ≈ -170.58 units.
- p-value < 0.0001, relationship is statistically significant.
- r^2 : 0.1703, model explains 17.03% of the variability in crime rate.
- F-statistic: 16.83 (p < 0.0001), confirming overall regression is statistically significant.

Linear Regression of Crime Rate on High School Diploma Percentage

The REG Procedure
Model: MODEL1
Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93462942	93462942	16.83	<.0001
Error	82	455273165	5552112		
Corrected Total	83	548736108			

Root MSE	2356.29195	R-Square	0.1703
Dependent Mean	7111.20238	Adj R-Sq	0.1602
Coeff Var	33.13493		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	20518	3277.64269	6.26	<.0001
hs_diploma_pct	1	-170.57519	41.57433	-4.10	<.0001

Hence, ↑ education levels are significantly associated with ↓ crime rate.

(c) Plot the residuals versus the explanatory variable and briefly describe the plot noting any unusual patterns or points.

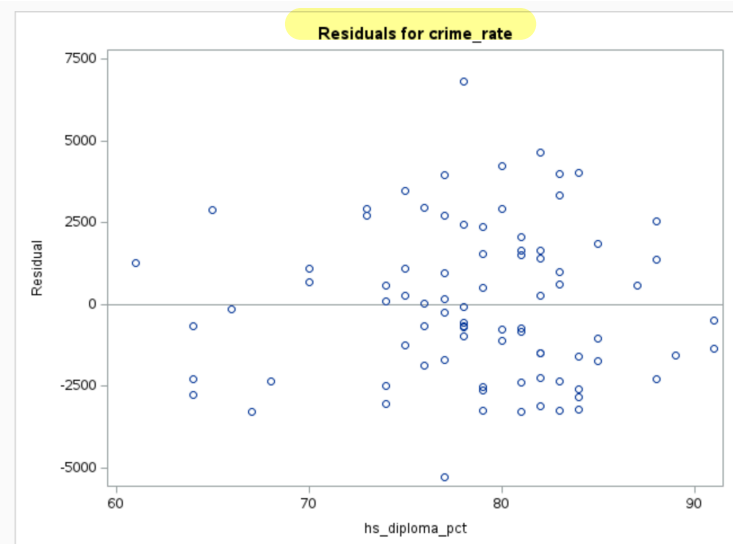
```
/* Run the regression and output residuals */
proc reg data=crime;
    model crime_rate = hs_diploma_pct;
    output out=resid_data r=residuals;
run;
quit;

/* Plot residuals vs explanatory variable */
proc sgplot data=resid_data;
    scatter x=hs_diploma_pct y=residuals;
    refline 0 / axis=y lineattrs=(color=red);
    xaxis label="High School Diploma Percentage";
    yaxis label="Residuals";
    title "Residuals vs. High School Diploma Percentage";
run;
```

- Residuals appear to be randomly scattered around zero with no systematic pattern: linearity ✓

- Although slightly ↑ of large residuals in positive direction, no indicator of heteroscedasticity or non-linearity.
- Although residuals appear to be wider when ↓ explanatory variable, not too extreme.

- Overall, residual plot does not violate any regression assumptions.

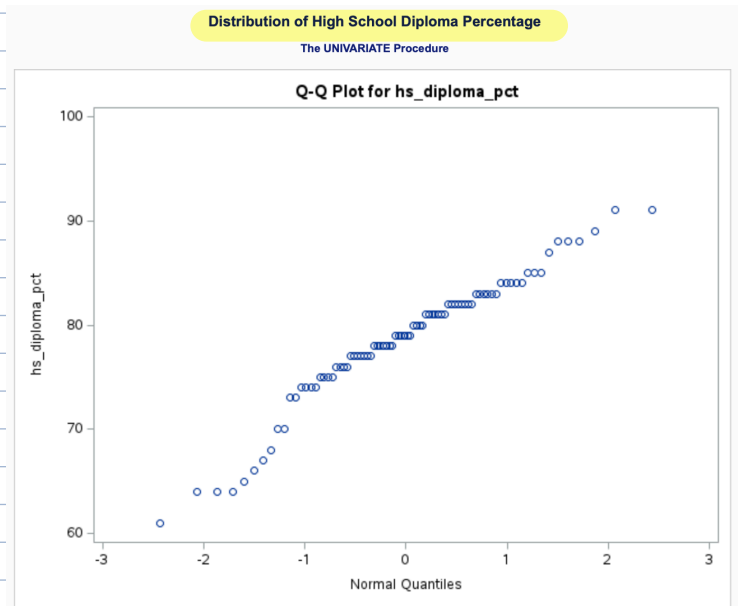
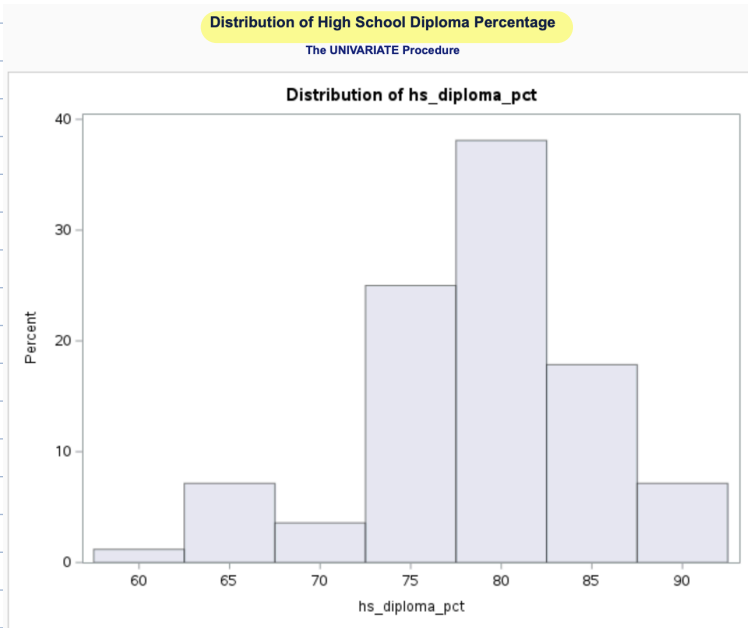


- (d) Examine the distribution of the residuals by getting a histogram and a normal probability plot of the residuals by using the HISTOGRAM and QQPLOT statements in PROC UNIVARIATE. What do you conclude?

```

91 proc univariate data=crime;
92     var hs_diploma_pct;
93     histogram;
94     qqplot;
95     title "Distribution of High School Diploma Percentage";
96 run;

```



- Unimodal and roughly symmetric, some indication of slight skewness or deviation of perfect normality.

- Most points fall along diagonal line, mostly normally distributed.
- Tails do deviate slightly off the line, indicating mild deviation of perfect normality.

- Both graphs seems reasonably normal, supporting validity of regression assumptions.
- While there are slight deviations in the tails in Q-Q, these are not severe and do not suggest major violations of normality.

5. (SAS Exercise) Use the **crime_rate** data described in KNNL Problem 1.28. Change the data set by changing the value of the crime rate for the last observation from 7582 to 758 (e.g., a typo). You can do this in a data step. For example,

```
DATA a2; SET a1; IF _n_ EQ 84 THEN y=758;
```

An alternative is to simply edit the data file.

- (a) Make a table comparing the results of this analysis with the results of the analysis of the original data. Include in the table the following: fitted equation, t-test for the slope, with standard error and p -value, R^2 , and the estimate of σ^2 . Briefly summarize the differences.

(SAS Code in 5b.)

Original:


The REG Procedure
Model: MODEL1
Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93462942	93462942	16.83	<.0001
Error	82	455273165	5552112		
Corrected Total	83	548736108			

Root MSE	2356.29195	R-Square	0.1703
Dependent Mean	7111.20238	Adj R-Sq	0.1602
Coeff Var	33.13493		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	20518	3277.64269	6.26	<.0001
hs_diploma_pct	1	-170.57519	41.57433	-4.10	<.0001

Modified:

The REG Procedure
 Model: MODEL1
 Dependent Variable: crime_rate

Number of Observations Read	84
Number of Observations Used	84

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	87518841	87518841	14.33	0.0003
Error	82	500804428	6107371		
Corrected Total	83	588323269			

Root MSE	2471.30959	R-Square	0.1488
Dependent Mean	7029.96429	Adj R-Sq	0.1384
Coeff Var	35.15394		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	20003	3437.63420	5.82	<.0001
hs_diploma_pct	1	-165.06193	43.60369	-3.79	0.0003

	Original	Modified (5b)
Fitted Equation	$\hat{y} = 20518 - 170.58(\text{hs_diploma_pct})$	$\hat{y} = 20003 - 165.06(\text{hs_diploma_pct})$
$\hat{\beta}_1$	-170.58	-165.06
SE of $\hat{\beta}_1$	41.57	43.60
t-statistic of $\hat{\beta}_1$	-4.10	-3.79
p-value of $\hat{\beta}_1$	<0.0001	0.0003
R^2	0.1703	0.1488
$\hat{\sigma}^2$ (MSE)	5552112	6107371

↓

- ↓ in magnitude of $\hat{\beta}_1$, slightly weakening relationship.
- ↑ SE, ↓ t-statistic, indicating ↓ precision.
- ↓ R^2 , indicating model explains ↓ variation.
- ↑ $\hat{\sigma}^2$, suggesting ↓ model fit.

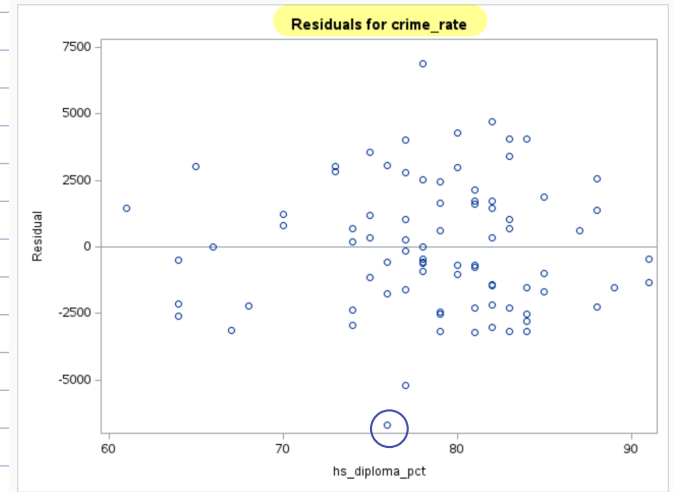
Hence, Modified Model ↓ strength and precision of the regression.

(b) Repeat parts (c) and (d) from the previous problem for this altered data set analysis and summarize how these plots help you to detect the unusual observation.

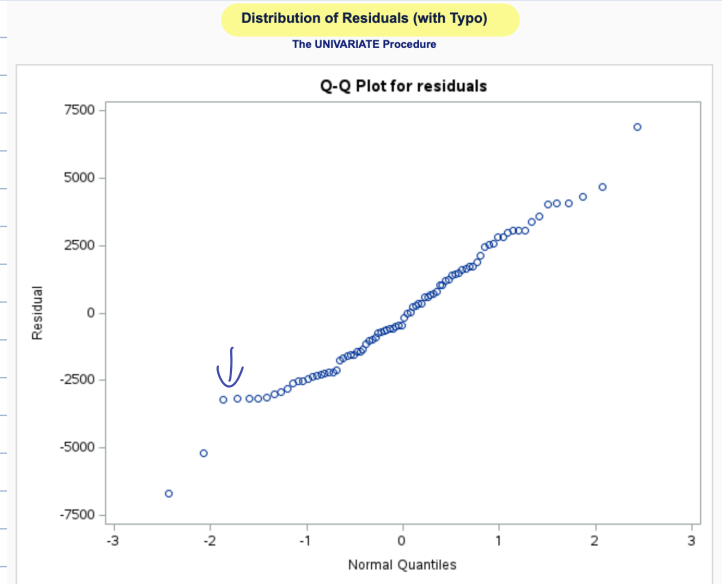
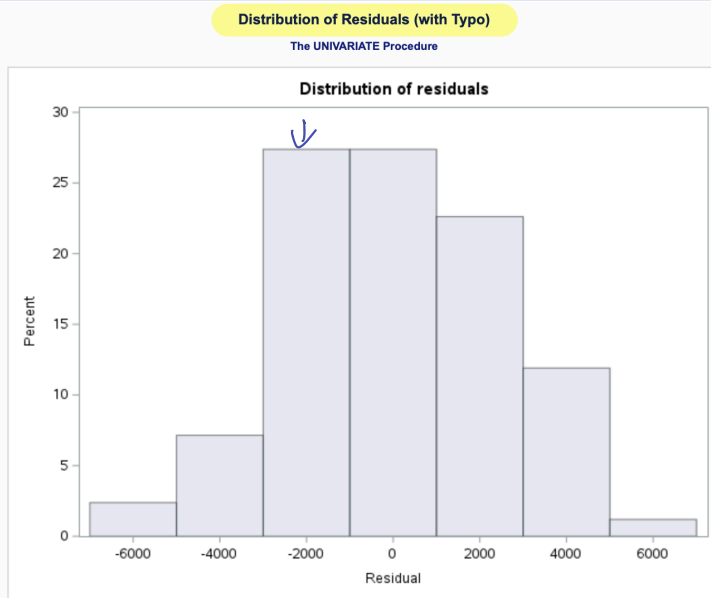
```

91 /* Step 2: Modify the last observation (observation 84) */
92 data crime_type;
93   set crime;
94   if _N_ = 84 then crime_rate = 758;
95 run;
96
97 /* Step 3: Run regression and output residuals */
98 proc reg data=crime_type;
99   model crime_rate = hs_diploma_pct;
100   output out=resid_typeo r=residuals;
101 run;
102 quit;
103
104 /* Step 4: Residual plot */
105 proc sgplot data=resid_typeo;
106   scatter x=hs_diploma_pct y=residuals;
107   refline 0 / axis=y lineattrs=(color=red);
108   xaxis label="High School Diploma Percentage";
109   yaxis label="Residuals";
110   title "Residuals vs. High School Diploma Percentage (with Typo)";
111 run;
112
113 /* Step 5: Histogram and QQ plot of residuals */
114 proc univariate data=resid_typeo;
115   var residuals;
116   histogram;
117   qqplot;
118   title "Distribution of Residuals (with Typo)";
119 run;

```



- One extreme outlier at $hs_diploma_pct = 76$.
- Modification at $N=84$ caused a large negative 'e'.



- No longer "bell-shaped".
- Skewed due to single error.
- Deviation from diagonal line at lower tail, violating normality.

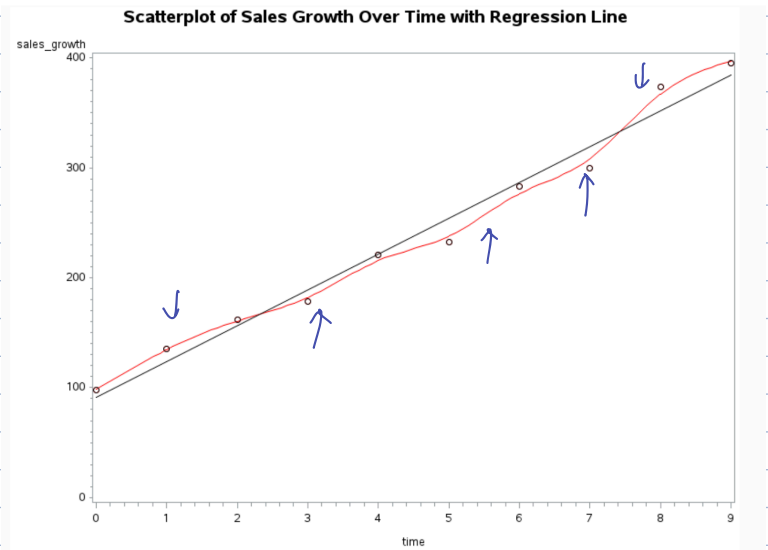
Hence, Modified Model ↓ strength and precision of the regression by:

- ↑ 'e' spread
- ↓ model fit (↓ R^2)
- Violate assumptions of normality and constant variance.

6. (SAS Exercise) Use the **sales growth** data described in KNNL Problem 3.17.

(a) Generate a scatterplot of the data and discuss the appropriateness of using a linear regression model.

```
3 data sales;
4   input sales_growth time;
5   datalines;
6   98.0 0.0
7   135.0 1.0
8   162.0 2.0
9   178.0 3.0
10  221.0 4.0
11  232.0 5.0
12  283.0 6.0
13  300.0 7.0
14  374.0 8.0
15  395.0 9.0
16 ;
17 run;
18
19 symbol1 v=circle i=sm50 c=red; /* scatterplot with smooth curve */
20 symbol2 v=circle i=rl c=black; /* regression line */
21 proc gplot data=sales;
22   plot sales_growth*time=1 sales_growth*time=2 / overlay;
23   title "Scatterplot of Sales Growth Over Time with Regression Line";
24 run;
25 quit;
```



- Sales Growth against Time shows clear upward trend.
- Points don't lie close to regression line, with tight 95% P.I, indicating inconsistency.
- Presence of non-linear relationship between Sales Growth and Time.

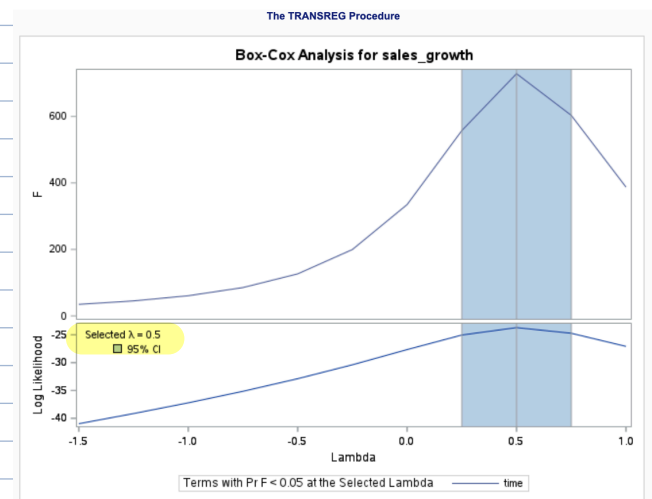
Hence, a Linear Regression Model may not be appropriate for modeling Sales Growth against Time without transformation.

(b) Using PROC TRANSREG, which power transformation of Y (i.e., value of λ) is most appropriate to use here?

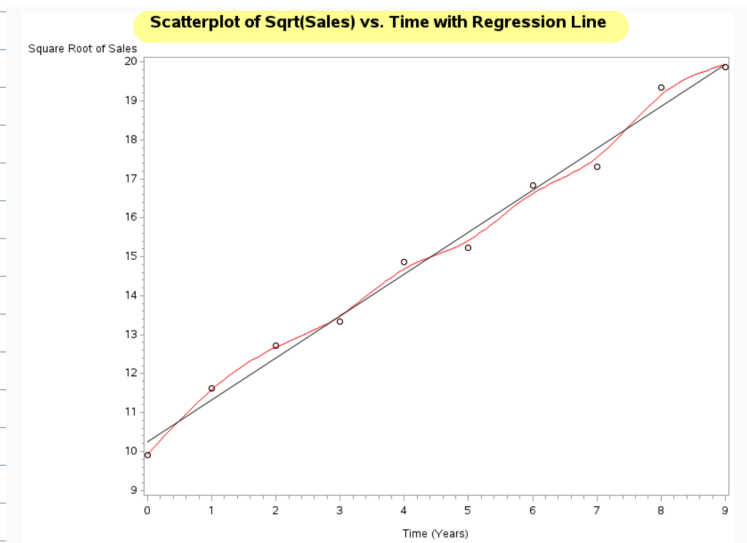
```
29 /* Box-Cox transformation using PROC TRANSREG */
30 proc transreg data=sales_growth;
31   model boxcox(sales_growth / convenient lambda=-1.5 to 1 by 0.25) = identity(time);
run;
```

parameter = λ

$\lambda = 0.5$, indicating square-root transformation of Y .



- (c) Apply this transformation of Y and generate a scatterplot. Again comment on the appropriateness of using a linear regression model.

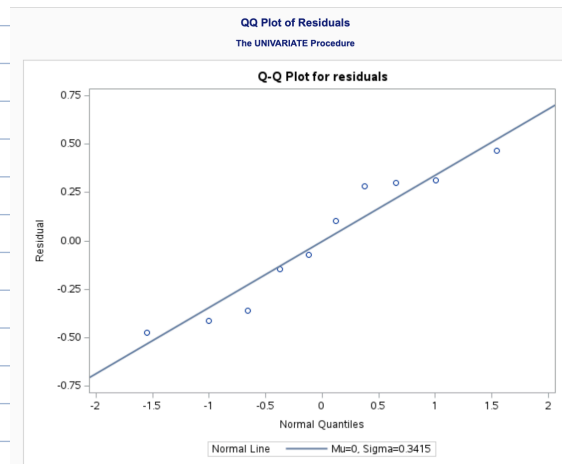


- After the square-root transformation, $\sqrt{Y}$ v.s. time, graph becomes more linear and ↓ curvature compared to the original plot.
- Points are closer to regression line, meaning variance is more stable, and this model is more appropriate.

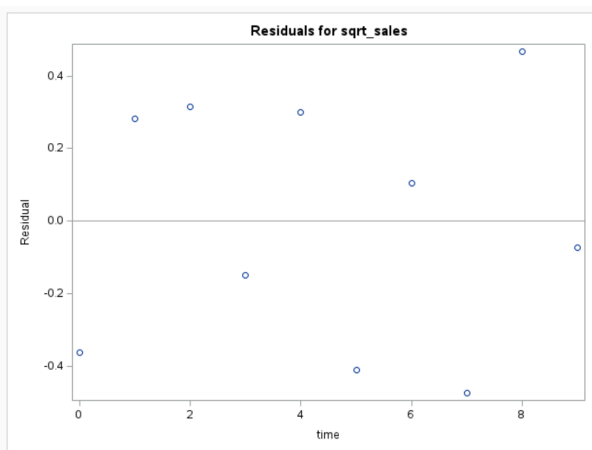
Hence, the square-root transformation to Y (Sales Growth) by $\lambda=0.5$ improves linearity, homogeneity, making a linear regression model more appropriate.

- (d) Run the regression model using the transformed data and generate a residual plot (using X or $\hat{Y}$) and a normal probability plot. What do the plots show?

```
18 proc reg data=a1;
19     model sqrt_sales = time;
20     output out=regout r=residuals p=predicted;
21 run;
22 quit;
23
24 symbol v=circle i=none c=blue;
25 proc gplot data=regout;
26     plot residuals*predicted;
27     title "Residuals vs. Predicted Values";
28 run;
29 quit;
30
31 proc univariate data=regout normal;
32     var residuals;
33     qqplot residuals / normal(mu=est sigma=est);
34     title "QQ Plot of Residuals";
35 run;
```



- Points align closer to the diagonal line, normality assumption satisfied.



- Both plots support of using a linear regression model after applying the square-root transformation.
- The linear model is a good fit for the transformed data.

- No clear pattern, linearity assumption satisfied.

(e) Express the estimated regression function in the original units.

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	10.26093	0.21290	48.20	<.0001
time	1	1.07629	0.03988	26.99	<.0001

$$(\sqrt{\text{Sales}})^2 = (10.26093 + 1.07629 (\text{time}))^2$$

$$\widehat{\text{Sales}} = (10.26093 + 1.07629 (\text{time}))^2$$